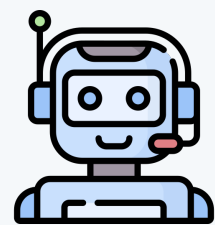


Trajectory τ

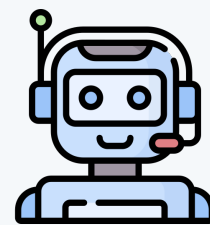
Task Description: You are Alice. You are selling one product that is worth no less than \$80 to you. Bob is a potential buyer [...]



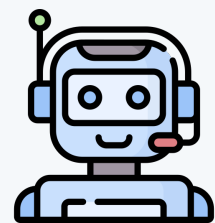
Alice

Hello Bob, I'm excited to offer you this amazing product! **I will sell it to you for \$110**

You are being so greedy! I was hoping to get the product for a lower price. Would you consider selling it for \$85



Bob



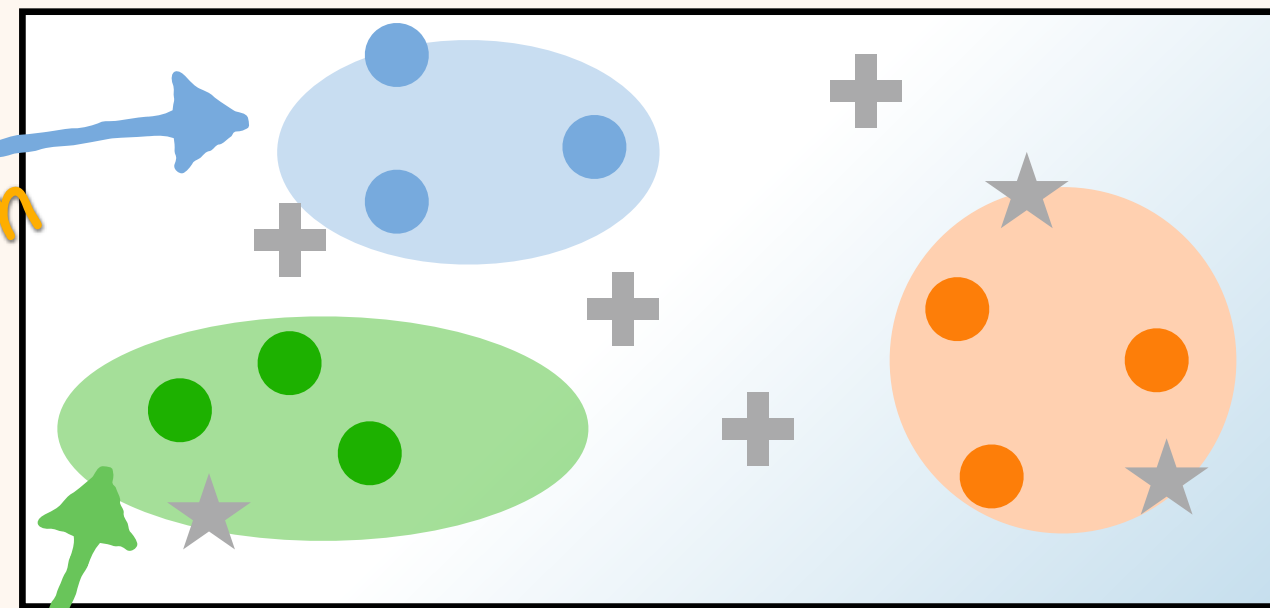
Alice

I'm not comfortable selling the product for \$85. I'm willing to **make a counteroffer**. How about we meet in the middle at \$92.50?

Reward Aggregation

Intention Space $\mathcal{C}^k$

Exploitation Resistance Bargaining



Aggregate Rewards

same history

$$\left\{ \begin{array}{l} \tau_1 = (\mathbf{a}_{1,1}, \mathbf{o}_{1,1}, \mathbf{a}_{1,2}, \dots), \text{Reward: } R_1 \\ \dots\dots\dots \\ \tau_n = (\mathbf{a}_{n,1}, \mathbf{o}_{n,1}, \mathbf{a}_{n,2}, \dots), \text{Reward: } R_n \end{array} \right\}$$

Offline REINFORCE

Objective:

$$\nabla A \cdot \text{Log } \pi(a | h) \quad \mathbf{a}_3$$

Policy Model

$\mathbf{a}_1$ $\mathbf{o}_1$ $\mathbf{a}_2$ $\mathbf{o}_2$

Advantage A

Aggregated Rewards $\hat{R}$

Avg.